



LECTURE 2

PLANNING

COEN / ELEC 390

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Acknowledgement

- The content of these slides is based on Chapter 4 of the textbook :

Product Design and Development by K.T. Ulrich, S.D. Eppinger
McGraw Hill Irwin

The Product Planning Process

- The product plan identifies the portfolio of products to be developed by the organization and the timing of their introduction to the market.
- The product plan is regularly updated to reflect changes in the competitive environment, technology and information on the success of existing products.
- Product planning decisions generally involve the senior management of the organization.

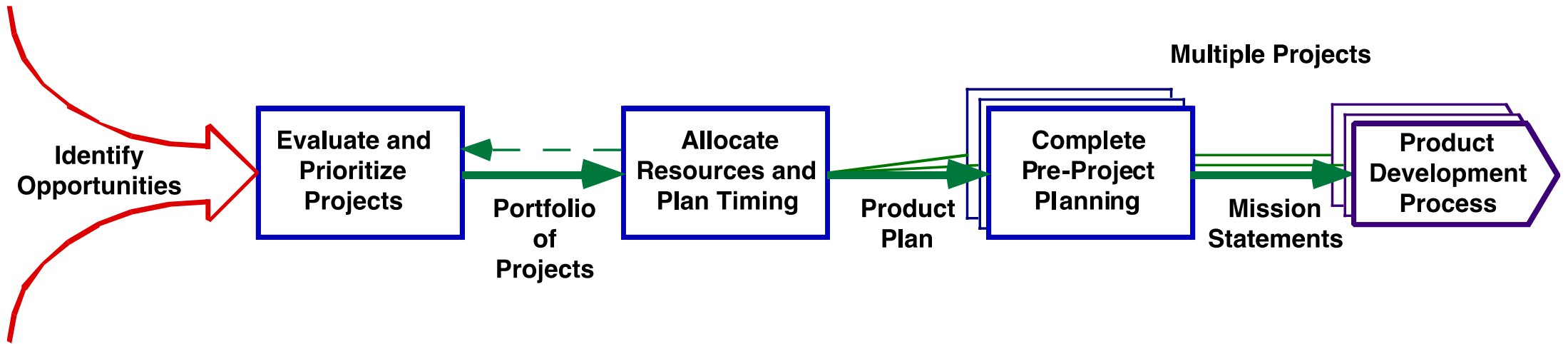
Importance of Planning

- Organizations that do not engage in careful planning are prone to a number of inefficiencies, including:
 - Inadequate coverage of target markets
 - Inappropriate timing of market introductions of products
 - Inadequate alignment between development capacity and the number of planned projects
 - Inappropriate distribution of resources, with projects exhibiting both overstaffing and understaffing

Types of product development projects

- New product platform
- Derivatives of existing product platforms
- Incremental improvements to existing products
- Fundamentally new products

The Process



Step 1: Identify Opportunities

- Ideas for new products or features of products may come from several resources, including:
 - Marketing and sales personnel
 - Research & technology development organizations
 - Current product development teams
 - Manufacturing and operations organizations
 - Current or potential customers
 - Third parties such as suppliers, inventors and business partners

Product Development Opportunities

- The identification of product development opportunities is closely related to identifying customer needs. Some proactive approaches include:
 - Document the frustrations and complaints of current customers
 - Interview lead users with a focus on their desired innovations
 - Follow trends in lifestyles, demographics, and technology
 - Study competitor products in detail.
 - Track the status of emerging technologies

Continued

- For each promising opportunity write an “**opportunity statement**”, a short coherent statement describing the opportunity.
- Example of an opportunity statement.

“Develop a new black and white (B&W), digital, networkable, document center platform for the office market, including scanning, storage, fax, distribution and printing capabilities.”

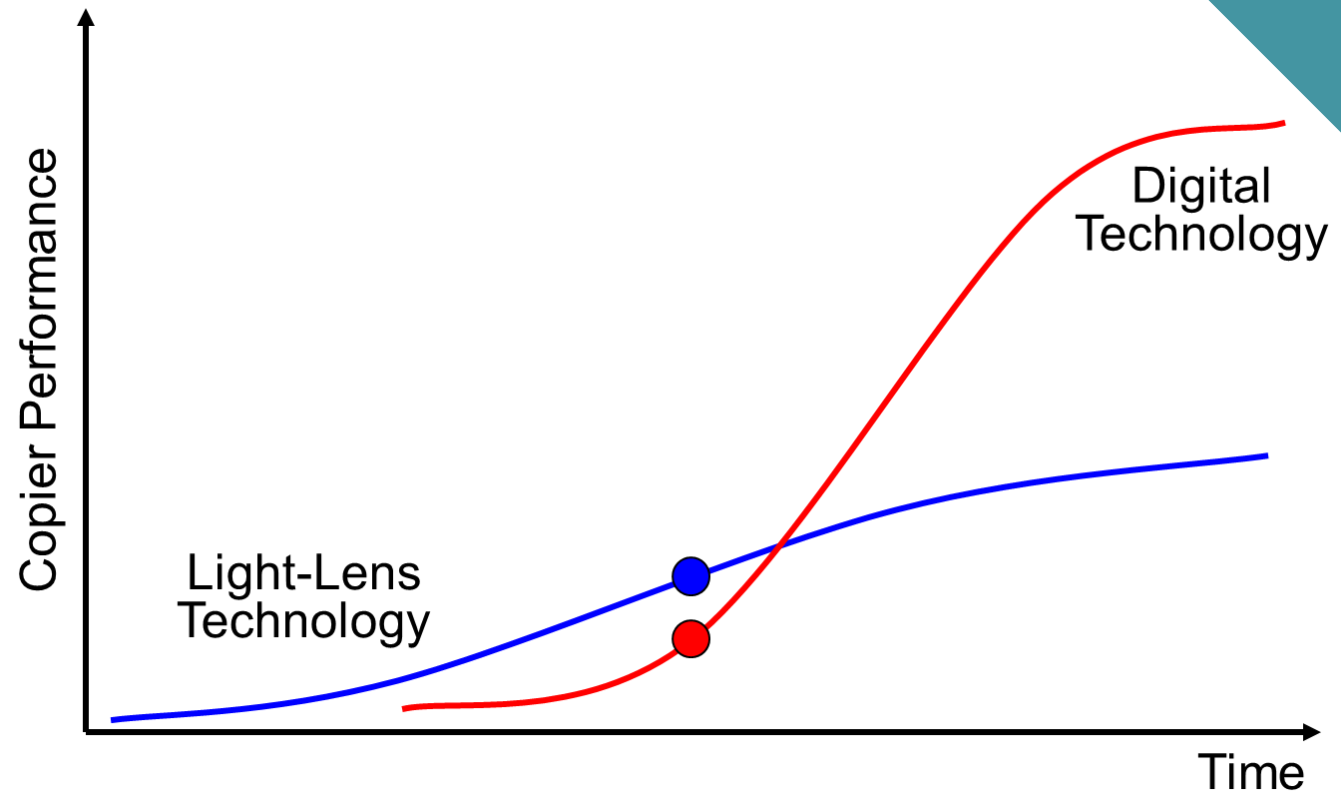
- **For your project, the opportunity statement should clearly state the customers you are aiming at.**
- Opportunity management systems exist.

Step 2: Evaluate & Prioritize Projects

- Four basic perspectives are useful in evaluating and prioritizing opportunities:
 - Competitive Strategy: The firm's basic approach to markets and product with respect to its competitors.
 - Market Segmentation: Different ways to categorize customers into segments.
 - Technological Trajectories: In technology-intensive business, a key product planning decision is when to adopt a new basic technology in a product line.
 - Product Platform Planning: An effective platform can allow a variety of derivative products to be created more rapidly and easily.

Technology S-Curve

- Technology S-curves are a conceptual tool to help think about technology trajectory.



Criteria to evaluate product opportunities

- Market size(units/year x average price)
- Market growth (percent per year)
- Competitive intensity (no. and strength of competitors)
- Depth of firm's existing market knowledge
- Depth of firm's existing technological knowhow
- Fit with other existing/future products
- Fit with existing capabilities
- Potential for patents and other barriers to competition
- Existence of a product champion within firm

Evaluate and Prioritize Projects

- For your project (adapted from fundamentally new product)
 - Identify market
 - Identify market size
 - Identify Competitors and their strengths
 - Consider team's knowledge of market
 - Consider team's knowledge of technology
 - Your access to customers for interviews.
- Evaluate under these and rank your product opportunities.

Step 3 Resource Allocation and Scheduling.

- Many organizations take on an excessive number of development projects, without due consideration of the development resources.
- The estimation of resources required on a monthly or quarterly basis compels the organization to confront the reality of limited resources.
- In instances where resource utilization exceeds 100%, the necessity arises to drop some projects.

Continued

- Project Timing. Sometimes this is called pipeline management. Factors to consider include:
 - Timing of product introduction
 - Technology readiness
 - Market readiness
 - Competition

Step 4: Complete Pre-Project Planning

- Provide clear guidance for the product development team by formulating a detailed definition of the target market and of the assumptions under which the development team will operate.
- These decisions are captured in a **Mission Statement**, which includes:
 - Brief (1-sentence) product description
 - Benefit proposition (why would a customer buy your product)
 - Key business goals (goal for cost, price)
 - Target market
 - Assumptions & constraints
 - Stakeholders (using customer, buying customer, distributor? Etc)

Step 5: Reflect on the Results of the Process

- Is the opportunity funnel collecting diverse exciting ideas?
- Does the product plan support the firm's competitive strategy?
- Does the product plan address the most important current opportunities?
- Are the development resources allotted sufficient?
- Have creative means of leveraging limited resources been considered?
- Does the core team accept the mission statement?
- Are the various elements of the mission statement consistent?
- Are the assumptions correct and accurate?
- How can the product planning process be improved?